

I deals in $k[x]$ ($k = \text{field}$)

Defⁿ : R : Commutative ring

The ideal generated by a is

$$(a) = \{ra : r \in R\}$$

Any such ideal will be called a principal ideal.

Theorem k : field

(i) Every ideal $I \leq k[x]$

is principal

i.e. $I = (f(x))$

for $f(x) \in k[x]$

(2) $K[x]/I$ is a field
if and only if

$I = (f(x))$
with $f(x)$ irreducible

Corollary (1):

Every finite field is
isomorphic to $\mathbb{F}_p[x]/(f(x))$

Where :

- p is a prime

- $f(x) \in \mathbb{F}_p[x]$

is an irreducible
polynomial

Corollary (2)

k : finite field

$$|k| = p^d, \quad p: \text{prime} \\ d \geq 1.$$

P. 1:

$$\left| \frac{\mathbb{F}_p[x]}{(f(x))} \right| = \left| \left\{ a_{d-1}x^{d-1} + \dots + a_1x + a_0 \right\} : a_i \in \mathbb{F}_p \right|$$

$d = \deg f(x)$

has p^d elements

Applications of Bezout

(1) $f(x) \in k[x]$ irreducible

$$f(x) \mid g(x)h(x)$$

$$\Rightarrow f(x) \mid g(x) \quad \underline{\text{or}} \quad f(x) \mid h(x)$$

Pf: Suppose $f(x) \nmid g(x)$

$$\text{Then } 1 = s(x)f(x) + t(x)g(x) \\ \perp \cdot h(x)$$

$$h(x) = s(x)f(x)h(x) \\ + t(x)g(x)h(x)$$

$f(x)$ divides RHS

(2) (Unique factorization)

$$a(x) \in K[x], \quad a(x) \neq 0$$

Then

$$a(x) = f_1(x)f_2(x) \cdots f_m(x)$$

where $\bullet f_i(x)$ are irreducible

• They are determined uniquely up to permutation and up to multiplying by constants

Pf:

$$f_1(x) f_2(x) \dots f_{\underline{m}}(x)$$

$$= g_1(x) g_2(x) \dots g_n(x)$$

Apply property (1) with the irred. poly. $f_1(x)$

$$\Rightarrow f_1(x) \mid g_r(x) \text{ for some } r$$

But since $g_r(x)$ is also irreducible, $g_r(x) = c f_1(x)$ for $c \in K^\times$.

$\Rightarrow$ We can cancel $f_1(x)$ from both sides.

Uniqueness is now shown by induction on m .

E.g.

$$(x^2+1)^2 = \left(\frac{1}{2}x^2 + \frac{1}{2}\right)(2x^2+2) \\ \in \mathbb{R}[x].$$

(3) Pt. of ① of Theorem

$$I \leq k[x]$$

$$\text{If } I = \boxed{(0)} = \{0\} \checkmark$$

If $I \neq (0)$, pick

$f(x) \in I$ of smallest possible degree

$$\underline{\text{Claim:}} \quad (f(x)) \stackrel{??}{=} \underbrace{I}_{\substack{\psi \\ \cap(x)}}$$

$$\underbrace{g(x)}_{\substack{\uparrow \\ I}} = \underbrace{q(x)f(x)}_{\substack{\uparrow \\ I}} + \underbrace{r(x)}_{\deg r(x) < \deg f(x)} \in I$$

$$\Rightarrow r(x) = 0$$

$$\Rightarrow a(x) \in (f(x)).$$

(4) Pf. of (2) of Theorem:

$K[x]/I$ is a field

$$(\Rightarrow) I = (f(x)), \quad f(x) : \text{irred.}$$

Pf. . If $K[x]/I = K[x]/(f(x))$

is a field, then

$f(x)$ is irreducible

[Same proof as in HW 1]

* If $f(x)$ is irreducible,

then $K[x]/(f(x))$ is a field.

$$0 \neq a(x) + (f(x)) \in k[x]/(f(x))$$

Bezout: $1 = \boxed{s(x)f(x)} + t(x)a(x)$
 $\exists s(x), t(x)$

$$\Rightarrow 1 \equiv t(x)a(x) \pmod{f(x)}$$

$$\text{[i.e. } 1 + (f(x)) = t(x)a(x) + (f(x)) \text{]} \\ \in k[x]/(f(x))$$

$$\Rightarrow t(x) + (f(x)) \text{ is the} \\ \text{inverse for } a(x) + (f(x)).$$

Proposition

R : comm. ring

$I \subseteq R$: subgroup

TFAT:

① I is an ideal

② $R \xrightarrow{\pi} R/I$ is

a homomorphism of rings

(3) $\exists \varphi: R \longrightarrow R'$
ring homomorphism
s.t. $I = \ker \varphi$

Remark: The fact that

any ring homomorphism

$$k \longrightarrow R$$

is injective

$(\Rightarrow)$ the only proper ideal
of k is (0) .

Remark: $I \leq R$: ideal

TFAE: (1) $I = R$

(2) $1 \in I$

(3) $\exists a \in I$

s.t. a is
invertible (w.r.t
multiplication)